AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I163*

Team Name: *Sirius*

Domain: *Healthcare, Biotechnology and Human Well-being*

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Date: 11th October 2026

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Problem Statement

"Cancer treatment often damages healthy cells along with tumor cells, causing unwanted side effects. The key challenge is delivering drugs specifically to the tumor, with precise, controlled and timely release, while minimizing exposure to healthy tissue"

Problem Understanding

- **Primary users**: Cancer patients receiving drug-based treatment, especially those whose treatment is limited by toxicity.
- **Cause**:The fundamental problem in conventional chemotherapy stems from a lack of selectivity and control. Traditional chemotherapy drugs are systemic poisons designed to attack all rapidly dividing cells. Because they cannot differentiate between a cancer cell and a healthy cell.
- **Effect on Users**: Cancer patients may experience side effects when therapeutic drugs affect healthy tissues, impacting their quality of life and treatment experience.
- **Effect on Stakeholders**: Healthcare providers face challenges in managing treatment-related side effects, while hospitals, pharmaceutical companies, and researchers need safer, more effective drug-delivery technologies. These challenges also create a need for innovations that improve treatment precision, safety, and patient outcomes.

Existing Gaps & Challenges

- **Premature Drug Release and Logic Failure**: The micro-capsule may release its payload before reaching the tumor due to leakage, unstable materials, or unintended activation by non-target biological signals. This could reduce targeting accuracy and expose healthy tissues to the drug.
- **Biological Barriers and Immune Clearance**: Blood proteins may form a coating around the capsule and interfere with its sensing or targeting functions. Additionally, immune cells in organs such as the liver and spleen may remove the capsule before it reaches the tumor.
- **Limited Tumor Penetration and Incomplete Drug Release**: Irregular tumor blood vessels and dense surrounding tissue may restrict the capsule's movement into deeper tumor regions. Even after activation, some of the drug may remain trapped inside the capsule.

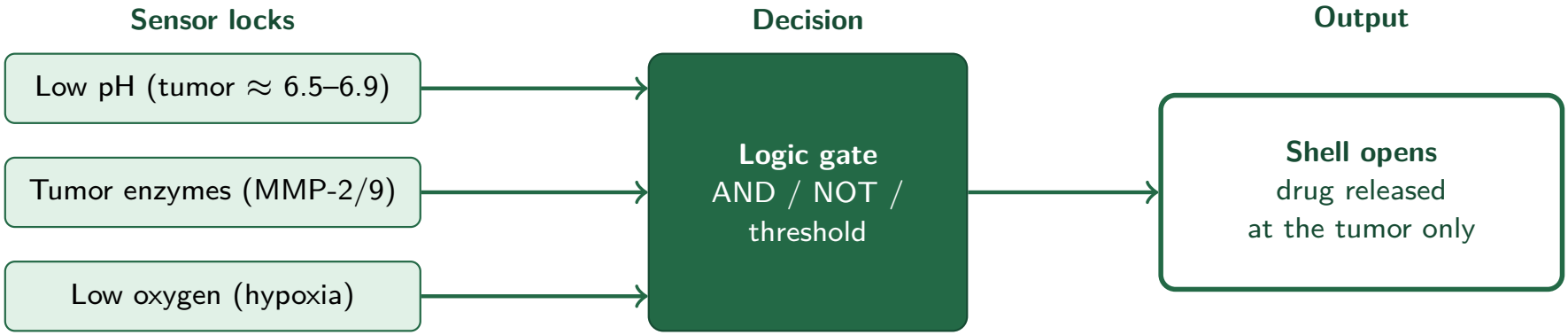
Our Proposed Solution

- **Core idea**: The Smart Tumor-Targeted Micro-Capsule Logic Gate is designed to deliver drugs more selectively to tumor tissues. It detects biological signals, such as a tumor-associated biomarker and acidic pH, and processes them through a logic-gate mechanism. When the required conditions are satisfied, the capsule triggers controlled drug release; otherwise, it aims to limit premature release. This approach could help reduce unnecessary drug exposure of healthy tissues.

- **How it works**: The capsule checks three tumor signals: acidity, tumor enzymes and low oxygen. The outer OR gate detects tumor-associated signals, the middle AND gate confirms multiple conditions, and the inner oxygen-responsive NOT gate helps regulate drug release under low-oxygen conditions.

- **Main value**: Aims to improve targeted drug delivery, reduce unintended exposure of healthy tissues, and potentially minimise treatment-related side effects.

Technical Approach



Capsules stay locked in blood. At the tumor, each lock reads one signal and the gate opens the shell only for the chosen combination.

Tech Stack (planned)

Materials: polymer or silica shell, cleavable linkers, doxorubicin (model drug)
Biology: Cancer and normal cell lines, 3D spheroids, plate-reader release assays
Characterization: particle size and zeta potential, TEM/SEM, HPLC
Modeling: Python or MATLAB (release kinetics), COMSOL (drug diffusion)

Innovation & Uniqueness

- The innovation lies in integrating AND, NOT, and OR logic gates into a smart tumor-targeted micro-capsule concept. Instead of responding to a single biological signal, the system uses combinations of signals to regulate drug release.
- Tumors can have complex and variable biological environments. Using multiple signals and logic-based decisions could help distinguish conditions suitable for drug release and reduce premature or unnecessary release in healthy tissues.
- The key differentiator is intelligent, multi-signal decision-making at the drug-delivery level. The AND, NOT, and OR gates provide different rules for controlling release, creating a potentially adaptable platform for more selective cancer therapy.

Feasibility & Scalability

- **Feasibility:** Builds on established research in responsive biomaterials and controlled drug delivery. Logic-gated release can initially be demonstrated using a safe model payload. A proof-of-concept prototype is potentially achievable with laboratory support; clinical development requires significant investment. Biological selectivity, safety, and therapeutic efficacy remain to be demonstrated.

- **Scalability:** Production could potentially scale using reproducible microencapsulation techniques and automated manufacturing. Maintaining consistent capsule quality, reliable logic-gate responses, and controlled drug release at large volumes remains a challenge.

- **Risks:** Potential risks include unintended drug release, false signal detection, immune reactions, and toxicity from capsule materials. Long-term safety, effectiveness, and clinical suitability must be established through rigorous testing before human use.

Use Cases

- **Use case 1: Tumor-Triggered Drug Release**
  - User: Cancer patients and oncologists.
  - Scenario: Micro-capsules are designed to release their drug payload when they encounter specific chemical signals associated with the tumor environment.
  - Expected Outcome: Drug release for tumor-associated condition

- **Use case 2: Multi-Signal Recognition**
  - User: Cancer researchers.
  - Scenario: A biological logic gate is designed to activate only when two selected tumor-associated signals are detected together.
  - Expected Outcome: Potentially greater selectivity than relying on a single signal, if the selected signals reliably distinguish tumor tissue.

- **Use case 3: Controlled Drug-Release Rate**
  - User: Pharmaceutical researchers.
  - Scenario: Researchers investigate capsule materials and designs that control how quickly a drug is released after activation.
  - Expected Outcome: A more predictable drug-release profile.

Use Cases

- **Use case 4: Personalized Drug-Delivery Research**
  - User: Biomedical researchers.
  - Scenario: Researchers investigate adapting the capsule's signal-recognition mechanism to biological markers associated with different patients' tumors.
  - Expected Outcome: Supports research into patient-specific drug-delivery.

- **Use case 5: Combination Cancer Therapy**
  - User: Oncologists and pharmaceutical scientists.
  - Scenario: Investigation on whether a capsule could carry two therapeutic agents and release them according to a predefined trigger or sequence.
  - Expected Outcome: Helps explore coordinated delivery of multiple therapeutic agents.

- **Use case 6: Laboratory Evaluation of Tumor Selectivity**
  - User: Laboratory researchers and drug developers.
  - Scenario: Researchers compare drug release under simulated tumor-like conditions with release under healthy-tissue-like conditions.
  - Expected Outcome: Quantitative evidence of how a capsule responds to the intended conditions.

Impacts and Benefits

- **Benefit to the target users**
  Aims to improve drug delivery precision by targeting tumor-associated signals. May reduce unnecessary exposure of healthy tissues to cancer drugs. Could support more controlled drug release compared with non-targeted delivery approaches.

- **Social, economic, or environmental impact**
  Potential to improve the quality of cancer care and patient well-being. More precise delivery could potentially reduce drug wastage and treatment-related costs. Biodegradable materials could help reduce persistent material waste, depending on the capsule design.

- **Expected measurable outcome**
  Percentage of drug released under target-like versus non-target conditions. Drug-release time and the ability to control the release rate. Percentage of drug delivered to the intended target in experimental models.
  Biocompatibility and degradation rate of the capsule materials.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 03	Target 3.4	Good health and well being.	Aims to improve cancer drug delivery precision.
SDG 09	Target 9.5	Industry, Innovation and Infrastructure.	Promotes innovation and research in targeted drug delivery.
SDG 12	Target 12.5	Responsible consumption and production.	Explores the potential to reduce drug wastage and use biodegradable materials.

Individual Member Contribution

Member	Contribution or responsibility
Dharshini Karuppasamy	Team Leader, Research paper gathering, Latex coding, Github repository work.
Akarshana R	Latex coding, Content gathering, Github repository work.
Shivani Manikandan Jagatha	Research paper gathering, Content gathering, Github repository work.
Hashini M	Content gathering.

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IR2026-1051043